MATH 1342 – Elementary Statistics – Week 15: ANOVA and Inference for Slope and Correlation

Nov 30 – Dec 2 | 8.1 Analysis of Variance • 9.1 Inference for Slope and Correlation | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Identify when ANOVA is appropriate and state H_0 and H_a in symbols and words. (CLO 8) **2.** Explain what SSG and SSE measure and how the ratio of between-group to within-group variability gives evidence against H_0. (CLO 8) **3.** Complete an ANOVA table (df, MS, F) from given sums of squares, interpret the F statistic and p-value, and check the conditions. (CLO 8)	e-Text: 8.1 Analysis of Variance Interactive Lecture: 8.1 Analysis of Variance Slides: 8.1 Analysis of Variance StatKey – Theoretical Distributions (F)	Participation: ANOVA – cuckoo-egg example: build the ANOVA table from given SS, find F, decide	Online Homework: 8.1 Analysis of Variance (due Sun Dec 6, 11:59 PM)	WileyPLUS StatKey Blackboard
4. State the simple linear model, distinguish the population slope β_1 and correlation ρ from the sample estimates b_1 and r, and interpret the slope in context. (CLO 7) **5.** Use regression output to test H_0: $\beta_1 = 0$ (equivalently $\rho = 0$) with a t statistic on $n - 2$ degrees of freedom, state the conclusion in context, and explain why the slope and correlation tests give the same p-value. (CLO 7, CLO 8) **6.** Construct and interpret a confidence interval for the slope, interpret R^2 as the proportion of variability explained, and check the conditions with a residual plot. (CLO 6, CLO 7)	e-Text: 9.1 Inference for Slope and Correlation Interactive Lecture: 9.1 Inference for Slope and Correlation Slides: 9.1 Inference for Slope and Correlation	Participation: Inference for Slope and Correlation – SleepStudy: depression score vs. classes missed (test for correlation and slope, CI for slope, R^2); self-quiz: drinks per week vs. classes missed	Online Homework: 9.1 Inference for Slope and Correlation (due Sun Dec 6, 11:59 PM)	WileyPLUS StatKey Blackboard
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Course Evaluation (in START HERE) – complete before the end of the semester	

CLO 6 Describe and compute confidence intervals. **CLO 7** Solve linear regression and correlation problems. **CLO 8** Perform hypothesis testing using statistical methods.